Contracting with Simplified Models

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Abstract

A receiver contracts a sender to help her make a choice. The sender estimates a "model" mapping the receiver's choices to expected outcomes. The receiver does not know whether the sender estimated the model on data "locally" tailored to a specific choice, or "distantly" across many plausible choices. The sender's models are also *simplified*: they may understate or neglect tradeoffs the receiver thinks are important, or overstate tradeoffs she thinks are unimportant. I show that senders who overstate tradeoffs estimate models on "local" data, while those whose models neglect tradeoffs estimate models on "distant" data. The receiver has lower forecast error when provided a model estimated on "local" data, and thus is better off contracting with a sender who *overstates* tradeoffs. I characterize when senders gain from providing simplified models, and show that the optimal method of "oversimplification" involves choosing a fixed (but arbitrary) degree of modeling a tradeoff. Finally, I study competition for information provision between multiple senders. Senders spatially sort across the receiver's choice space: those who tend to underestimate tradeoffs locate themselves in the center, while those who overstate tradeoffs locate on the edge. When the number of potential senders is sufficiently high and costs of information provision are sufficiently low, the senders' "gains from simplicity" vanish, as the receiver is able to contract multiple senders and uncover the true data-generating process.

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1 Introduction

Consider an economist tasked with designing a carbon tax. While she understands the economic tradeoffs underlying taxation, she knows very little about the relevant science and engineering, such as the mapping between emissions and temperature, the long-run trajectory of climate change, or what sorts of manufacturing processes are most pollutive. So, she hires a climate scientist to help her. The scientist collects data, runs complex simulations, and synthesizes his findings into a *model* that illustrates the costs and benefits of different tax rates. The model succinctly maps choices to outcomes that are relevant for the economist.

Yet, learning via a *model* creates frictions. Because the model synthesizes complex data, simulations, and domain expertise, there is no sense in which the economist can substantively audit how the scientist fit the model. More broadly, the scientist and economist have different ways of approaching the problem. The scientist's model may take an overly *simplistic* view relative to the economist. He may mechanically overweight ecological spillovers the economist views as second-order, or neglect political backlash from high tax rates. The economist could seek additional opinions, but this is costly, and she may not necessarily learn more. Thus, while contracting is valuable, uncertainty about *how models are formed* shapes the quality of information.

This illustrative example reflects a broader set of economic interactions where a principal contracts an agent to provide expert analysis that guides her decisionmaking. A CEO may enlist a management consultant to guide workforce restructuring. An investor may hire a portfolio manager to inform investment decisions. A voter may ask a politician to justify a policy with evidence from a pilot. Or, a student may consult an AI agent for help with a mathematical proof. In all these cases, the agent does not present the principal with with a *model* — which is a synthesis of (potentially proprietary) data the agent may have access to and the agent's expertise. These models descirbe how the receiver's actions — layoff decisions, stock choices, or tax rates — map onto outcomes — profits, financial returns, or voter wellbeing.

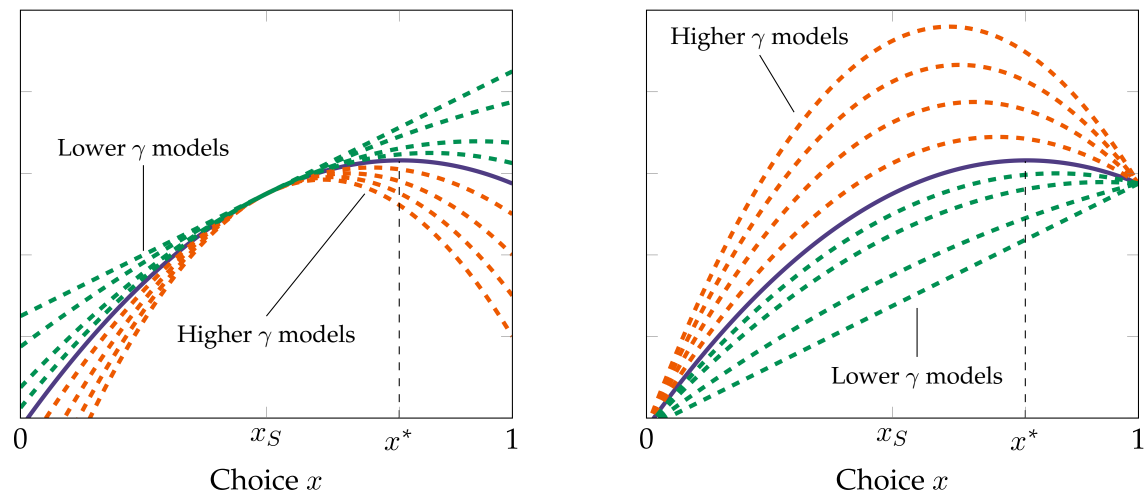
This paper theoretically studies frictions that emerge when contracting with models. A receiver (principal) needs to make a choice. She contracts a sender (agent) to provide information. The sender gathers data — which may also include his own experiences and expertise — and estimates a *model*: a data-generating process (DGP) mapping actions to outcomes. Consultants, for instance, routinely present clients with a series of "if-then" scenarios that describe how a firm's profits vary with the intensity of layoffs, or how investment returns depend on portfolio choices. However, the model may be *simplified* in that it neglects key tradeoffs, or emphasizes risks the receiver thinks are unimportant. Moreover, the receiver does not know much about how the model was estimated — and in particular whether it was "locally" estimated on a small set of choices or included information about all of the receiver's potential actions.

I address four broad questions. First, if the sender estimates a simplified model, how does the receiver learn about the true DGP, and how does this depend on beliefs about the *data* it was estimated on? Second, does the sender prefer to gather information "locally" about a small set of choices or "distantly" across all of the receiver's potential choices, and how does this depend on the class of models he can supply? Third, can a sender benefit from supplying models that deliberately understate or overstae tradeoffs? Finally, how are these distortions shaped by *competition* among senders — including market power, overhead costs, and the receiver's ability to consult multiple senders?

Model Overview A receiver — a CEO, investor, or voter — must choose an action x to maximize an outcome y — such as profits or the value of a public good. She believes that the relationship between x and y is given by a DGP $y = \alpha_0 + \alpha_1 x - \gamma x^2$ for $\gamma \geq 0$. This captures a tradeoff between a linear benefit and a convex cost, with γ representing the intensity of this tradeoff. However, she does not know the values of these parameters. She thus specifies an action x_S for the sender to investigate. The sender samples x values from a distribution and observes the corresponding y values, producing data (y, x). She may "locally" sample information about x values close to x_S, or "distantly" sample information about x values far from x_S. His sampling choice is unobserved by the receiver.

The sender then fits a model to the data. He considers similar models to the receiver, except that the set of γ values available to him is a subset of the receiver's, limiting his ability to model tradeoffs. Sender types are heterogeneous and differ in the range of γ values they can fit models to. We say a sender's models *understate* tradeoffs if the set of γ values they consider tend to be *lower* than the ones the receiver considers. They *overstate* tradeoffs if the γ values they consider tend to be *larger*. The receiver observes the sender's model, updates her beliefs over the true DGP, and makes her decision. The sender seeks to maximize the receiver's posterior expectation of y. For example, a portfolio manager may receive a cut of an investor's profits, a consultant's bonus may depend on his client's optimism, or a politician may want to maximize voter engagement.

Receiver Learning and Sender Sampling The first result is that receiver learning is shaped by the interaction of the sender's modeling limitations with his sampling strategy. The sender provides a model, shown in purple in the figure below. However, the receiver may still entertain alternative models. These plausible models may have higher or lower γ values. Under local sampling, the receiver's set of plausible models is tangent to the sender's model at $x = x_S$, as shown on the left. At x^* — where the sender's model is maximized — models with higher γ (orange) are below the receiver's model. Models with lower γ (green) are above the sender's model,. Under distant sampling, the set of plausible models shifts vertically, shown on the right. Because all models share the same sample mean, higher γ models lie above the sender's model at x^* and lower γ ones lie below.



(a) Plausible models under local sampling

(b) Plausible models under distant sampling

Figure 1: Plausible models when the sender's model understates tradeoffs

This leads to a second result: when the sender's models tend to *overstate* tradeoffs, he benefits from local sampling. In this case, any additional models the receiver entertains have lower γ than the sender's. Concretely, the green lines in the figure lie above the sender's model at x^* under local sampling, but below them under distant sampling. The sender's perception of the receiver's expected utility is higher in the former case. However, if the sender's models *understate* tradeoffs, he prefers distant sampling. The receiver's additional models have higher γ than the sender, lying above the sender's model under distant sampling but below under local sampling. A key corollary is that a sender restricted to *linear regression* models ($\gamma = 0$) always prefers distant sampling. This raises a tension between the interpretation of a regression as a "local linear approximation" to a data-generating process and its strategic use by a sender to manipulate a receiver's optimism.

Third, this interaction between model misspecification and sampling has implications for receiver welfare. Under distant sampling, the vertical dispersion across plausible models increases at x^*, so if the receiver values forecast accuracy, she prefers local sampling. Thus, the receiver learns the true DGP more precisely when the sender overstates tradeoffs and samples *locally*.

Gains from Oversimplification Fourth, a sender may gain from model oversimplification. I characterize when a "sophisticated" sender — who can fit any γ in the receiver's feasible set — benefits from the presence of "simple" sender types restricted to a tighter range of γ values. Upon observing a model, the receiver believes she received the correct model from the sophisticated sender or a misspecified model from the simple sender. I show that the sophisticated sender gains when the simple sender samples in a way such that any alternative models lie above the reported model. I show that if a "type-space designer" could choose an optimal distribution of simple sender types to maximize payoffs, he would have one sender for each γ value who can only fit models with that specific γ value. The key implication is that organizations benefit from deploying senders with a fixed—but arbitrary—way of modeling the tradeoff intensity γ.

Sender Competition The final set of results study competition between multiple senders who face overhead costs of providing a model. Receivers are heterogeneous in the actions they would like to receive information about. Senders specialize in providing information about different actions. If the action space is bounded, senders specializing in extreme actions must sample "less distantly," as their proximity to the boundary limits the variance of x values. Consequently, low-γ senders who understate tradeoffs and sample distantly concentrate in the center of the action space. Since distant and local samplers are each made worse off by being mistaken for the other, they never locate in similar regions, and high-γ senders who overstate tradeoffs concentrate near the boundary of the action space. Competition thus introduces *spatial sorting* across sender types.

If receivers value forecast accuracy those seeking information about interior actions face a tradeoff between learning accurately about less valuable actions and less accurately about more valuable ones. This tradeoff vanishes when overhead costs are sufficiently low and the market is sufficiently competitive. In this case, a "market for second opinions" emerges: a receiver contracts with two senders and combines their information to recover the true DGP. I show that while this erases the gains a "sophisticated" sender derives from the existence of "simple" senders, it benefits "simple" senders by sharpening the set of plausible models the receiver entertains.

Section 2 lays out the basic model, and I characterize learning in section 3. Section 4 studies gains from oversimplifying models and the design of an optimal type space for a sophisticated sender. Section 5 investigates how these distortions become change when considering competition between multiple senders. Section 6 concludes. The remainder of this section gives an overview of the literature.

Literature

2 Benchmark Model

We begin by laying out a game between a single receiver R (she) and a single sender S (he). R decides whether she would like to contract the services of a sender S. She specifies a choice for S to investigate. S collects data about this choice and provides a R a simple model describing that data. R's ability to commit S to a data sampling strategy or more complex models is limited. Taking this into account, R utilizes the information provided in S's model to make a decision. We abstract from prices and competition, but return to study these questions in Section 5.

2.1 Receiver and Models

A receiver R faces a set of choices X, where X is a closed and convex subset of $\mathbb{R}$. Each of these choices maps to a distribution of outcomes $y \in \mathbb{R}$. Ex-ante, R does not know the distribution of outcomes. However, she believes that the distribution of y given x can be described by a data-generating process $y = f_R(x)$ for a function $f_R : X \to \mathbb{R}$. We refer to f_R as a **model**. We denote $\mathcal{F}$ as the set of *models* f_R that R entertains. $\mathcal{F}$ is closed and convex. We make the following assumptions on $\mathcal{F}$.

Assumption 1. *For each $f_R \in \mathcal{F}$, we have*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2$$

for $(\alpha_0, \alpha_1) \in \mathbb{R}^2$ and $\gamma \in \mathbb{R}^+$.

Assumption 1 says that each of R's functions is concave, and can in particular be expressed as an affine function less a convex, quadratic loss. Notably, since $-\gamma x^2$ is concave, the functional form can also represent the consideration of diminishing returns in consumer utility, production, or any other concave function in shaping an outcome. We assume a quadratic form for tractability but show that the central intuitions generalize in section 3.5.

As a shorthand, we will denote the affine component of f_R as: $\ell(x; \alpha_0, \alpha_1) = \alpha_0 + \alpha_1 x$. We define as $\mathcal{L}$ the subset of functions $\mathcal{F}$ that are affine:

$$\mathcal{L} = \{f_S \in \mathcal{F} : f_S = \ell(x; \alpha_0, \alpha_1)\}$$

R has a prior p over the set of models $\mathcal{F}$. We make the following assumptions over the prior for tractability:

Assumption 2. *There exist densities p_α with support on $\mathbb{R}^2$ and p_γ with support on $[0, \overline{\gamma}]$ for $\overline{\gamma} > 0$ such that if $f_r(x) = \alpha_0 + \alpha_1 x - \gamma x^2$:*

$$p(f_r) = p_\alpha(\alpha_0, \alpha_1) \cdot p_\gamma(\gamma).$$

Moreover, $p_\alpha(\alpha_0, \alpha_1) = p_\alpha(\alpha_0', \alpha_1)$ for all $\alpha_0, \alpha_0' \in \mathbb{R}$.

TODO: fix this at x_S This assumption says that the prior over the potential values of Γ is independent of the affine component $\ell(x; \alpha_0, \alpha_1)$, and that R places the same prior mass on vertical shifts of each model. Throughout, we will use capitals P_α and P_γ to refer to the CDFs of these distributions.

Given a belief q over models, R's expected utility is given by:

$$U_q(x) = \mathbb{E}_q[f_R(x)].$$

Her goal is to choose x to maximize $U_q(x)$. The receiver values the expected value of the outcome y given her choice x. If R is an investor, she cares about how much money she can make by investing in a choice x. If R is a CEO restructuring her firm via layoffs, she cares about her decision's impact on profits. If R is a voter, she cares about the impact of a policy x on her wellbeing. We utilize these simpler preferences in our baseline model, but section 3.4 extends them to evaluate settings where R also values *forecast accuracy*, and thus knowing the correct model of the world.

2.2 Sender and Data

The receiver R contracts a sender S to provide her with information about the true data-generating process. The sender has a private type affecting the simplicity of the models she considers. The order of events is as follows.

1. Nature draws S's type.
2. R pre-specifies a choice $x_S \in X$.
3. S collects data about the choice x_S.
4. S provides a model to R that fits the data he collected. R updates her beliefs about the true data-generating process based on her beliefs about S's type and the model.
5. R makes a decision x^*. S receives utility. Uncertainty resolves, and R receives her utility.

We now provide more detail about each of these components.

Sender's Type A sender's type affects the set of models — and hence information — he is capable of providing to R. The key is that S is limited in how he can model the non-affine component of f_R. To this end, we denote the type space by $\Theta \subseteq \mathcal{P}[0, \overline{\gamma}]$ — i.e. each type $\theta \in \Theta$ corresponds to a subset of $[0, \overline{\gamma}]$. The sender's type is drawn from a distribution T_θ over Θ, which is known to R. Θ has the following properties.

Assumption 3. *For each $\theta \in \Theta$, $\theta \subseteq [0, \overline{\gamma}]$ is closed and convex. Moreover, for each $\theta \neq \theta'$, $\theta \cap \theta' = \varnothing$.*

Assumption 3 states that each type corresponds to an interval within $[0, \overline{\gamma}]$. This interval may be degenerate. We assume the intervals do not intersect for expositional purposes, but the case where they do not intersect is of particular interest, and we study it in section 4.

A type θ sender is able to provide R a model in the set F_θ, where

$$F_\theta := \{f_R \in \mathcal{F} : f_R = \beta_0 + \beta_1 x - \delta x^2, \delta \in \theta, (\beta_0, \beta_1) \in \mathbb{R}^2\}.$$

Intuitively, θ represents the *degree* to which a sender can model the *non-linear* component of a data-generating-process f_R. For example, suppose there is a single type $\theta = [0, \overline{\gamma}]$. Then, S is able to provide R any of the models in $\mathcal{F}_R$. Of particular interest is the case where $\theta = \{0\}$, in which case S can only provide information to R in the form of a "linear regression."

Sender's Data Given a pre-specified choice x_S, S chooses a data sampling strategy μ_S, where μ_S is a distribution defined over X. Intuitively, this means that S is choosing a distribution of choices in x to collect data on. R does not observe S's choice of μ_S, meaning data-collection suffers from a *lack of verifiability and commitment*.

Assumption 4. *μ_S is parametrized by $\mu_S(x_S, \sigma) = x_S + \sigma \cdot \epsilon$, where ϵ is mean-zero, symmetric, and $0 < \mathrm{Var}(\epsilon) < \infty$. Moreover, $\sigma \in [\underline{\sigma}, \overline{\sigma}]$ for $\underline{\sigma} > 0$ arbitrarily close to 0.*

Assumption 4 provides a convenient parametrization of μ_S as a distribution centered around x_S. σ as a parameter effectively controls the spread of the distribution. The constraint that $\sigma > 0$ is a normality assumption made to ensure learning is structured in equilibrium. We will refer to σ as a choice variable for S, and denote σ_θ the equilibrium choice of σ for a type θ sender.

Given this sampling strategy, S directly observes the joint distribution of (y, x), subject to observational noise. Formally, if the true data-generating-process is f_R^*, she observes a joint distribution $D(y, x)$, where the marginal distribution of x is given by $D_x = \mu_S$ and the conditional distirbution of y given x is given by $D_y(y|x) = f_R^*(x) + \eta$, where η is a mean-zero noise term with unknown variance. This is equivalent to S observing an infinite number of i.i.d. draws from the true data-generating-process.

Sender's Information With the sender's type and sampling strategy, in hand, we define the model S provides to R. We say that a model m **best-fits the data** with respect to θ and a data-sampling strategy μ_S if:

$$f_S = \arg\min_{f \in F_\theta} \mathbb{E}_{D(\mu_S)}\big[\big(y - f(x)\big)^2\big].$$

Our key assumption is that S provides R the model f_S that best-fits the data, given his type θ and μ_S. The existence of f_S is given by the Hilbert Projection Theorem. For example, if $F_\theta = \mathcal{F}$, then f_S will reperesent the true data-generating process. Or, if $F_\theta = \mathcal{L}$, S provides the best-fitting *linear regression model* to the data. Crucially, both θ and μ_S are unobserved by R, although R may be able to infer these objects in equilibrium.

Sender's Preferences We assume type θ sender has a prior p^θ over the models in $\mathcal{F}_\theta$. We assume that p^θ can decomposed into $p^\theta_\beta(\beta_0, \beta_1)$ and $p^\theta_\delta(\gamma)$ and accordingly satisfies Assumption 2. Suppose S provides a model $f_S \in \mathcal{F}_\theta$. Let x^* be the receiver's choice of $x \in X$ conditional on her posterior beliefs $q|f_S$. The sender's utility conditional on f_S is given by:

$$U_S(x^*) = \mathbb{E}_{q|f_S}[f_R(x^*)].$$

That is, conditional on providing a model f_S, S would like to maximize R's expected utility. These preferences lend themselves to several interpretations. For example:

- S is a portfolio manager who receives a fixed share of R's expected earnings. Then, he cares about the expected profits R can realize (and thus his payment).
- S is a consulting firm whose services a CEO R contracts. S may care about the expected gain in profits R receives either directly (e.g. because his "bonus" is proportional to R's gain in profits) or indirectly (e.g. because his reputation is tied to his ability to realize profits for those who contract his services).
- S is an office-motivated politician who provides information to a median voter R. The higher the median voter thinks y is, the more likely she is to turn out. If re-elected, the politician will implement R's preferred policy x^*.

2.3 Comments

The model assumes that R cannot verify information on two fronts: *data* and *functional form*. The lack of commitment to a specific dataset underscores a key feature of contracting with models. R knows S has access to useful data, but does not know exactly what that data looks like or how it is being used to provide her with a model. The clearest example is one where S's data is exclusive and proprietary — R cannot observe these data directly, and must take into account S's incentives and the information she is provided to make inferences about what the data might be. Portfolio managers and consulting firms may have access to past data on returns or cases they have worked on which they are unable to disclose directly, or politicians may have access to confidential data on past policy experiments collected by the public service. Another example is an AI model which detects high-level data patterns in a "black-boxed"' way that cannot be elucidated to a user (or even its designers).

While the receiver R may be unable to access these data for legal reasons, she also be unable (or unwilling) to expend organizational time and resources in collecting, understanding, and synthesizing complex data. Instead, she delegates the collection of data and its synthesis into a model to the sender, expediting her decision-making process in the interest of convenience, but potentially at the cost of information.

This is related to the second dimension of nonverifiability, which is that S may not necessarily fit the true data-generating-process in the way R would like. The sender's type constrains how he can model the true data-generating-process, which may be affected by technological feasibility or a difference in thinking about a problem the same way R does. For example, suppose S is a climate scientist and R is an economist. S provides climate modeling to help R assess the impact of a climate shock on the demand for electric vehicles. Even if S understands the concept of demand, there may be technical constraints on the sorts of parameters his advanced climate models can provide to R. In particular, because $-\gamma x^2$ represents a convex loss, model simplicity particularly constrains S's ability to model *tradeoffs*. Thus, S may be forced to provide a model that understates or overstates the degree of a tradeoff.

Many of the results will involve thinking about what happens when S can only provide linear regression models to R. This is motivated by the widespread usage of linear regression models in the business world, economics, and data-driven social science. We will be interested in understanding not only how we can learn about non-linear processes through linear models, but also how the use of linear models coupled with data sampling strategies can be used to manipulate beliefs about the effects of a policy or intervention. Moreover, we will contrast how the behavior of senders constrained to provide linear models may in particular differ from the behavior of senders constrained to provide other simplified (but nonlinear) models.

For the majority of the analysis, we focus on settings where the nonlinear component of f_R is quadratic. While this is done for tractability, section 3.4 discusses the sorts of assumptions necessary to generalize the logic to more general functional forms. Most of the results also generalize to multidimensional settings, and many of the proofs are identical, which we discuss in the same section.

3 Learning from Simple Models

3.1 Receiver's Learning and Behavior

We begin by characterizing receiver learning as a function of data-sampling strategies and S's type.

Receiver's Updating Suppose S provided a model of the form $y = \beta_0 + \beta_1 x - \delta x^2$. What models would R entertain? First, note that due to Assumption 3, R is able to infer θ from δ. We define $\phi(f_S; \theta, \mu_S)$ as the set of functions that could rationalize the data, given f_S best-fits with respect to F_S and μ_S:

$$\phi(f_S; \theta, \mu_S) = \left\{ f_R \in \mathcal{F} : f_S = \arg\min_{f \in \theta} \mathbb{E}_{\mu_S}\big[\big(f_R(x) - f(x)\big)^2\big] \right\}.$$

If $f_R \in \phi(f_S; \theta, \mu_S)$ We say that f_R **fits the data** with respect to f_S, θ, and μ_S if $f_R \in \phi(f_S; \theta, \mu_S)$. Since S's type does not constrain β_0, β_1, we necessarily have that for each $f_R \in \phi(f_S; \theta, \mu_S)$:

$$\frac{\partial}{\partial \beta_i} \mathbb{E}_{\mu_S}\big[\big(f_R(x) - f(x)\big)^2\big] = 0,$$

which yields the following moment conditions familiar from ordinary-least-squares (OLS):

$$\mathbb{E}_{\mu_S}\big[f_R(x) - \beta_0 - \beta_1 x - \delta x^2\big] = 0 \tag{1}$$

$$\mathbb{E}_{\mu_S}\big[x\big(f_R(x) - \beta_0 - \beta_1 x - \delta x^2\big)\big] = 0 \tag{2}$$

Equation (1) is the moment condition with respect to the intercept, and states that on average, the difference between f_R and f_S must average to zero. Equation (2) can be combined with equation

(1) to read that the covariance between x and $f_R(x) - \beta_0 - \beta_1 x - \delta x^2$ is zero. Both expectations are taken with respect to μ_S, which is S's sampling strategy.

Additionally, we also have a moment condition corresponding to δ:

$$\mathbb{E}_{\mu_S}\left[x^2\left(f_R(x) - \beta_0 - \beta_1 x - \delta x^2\right)\right] \tag{3}$$

which may be zero or nonzero, depending on S's type.

The following proposition characterizes the set of functions that R entertains given f_S, θ, and μ_S.

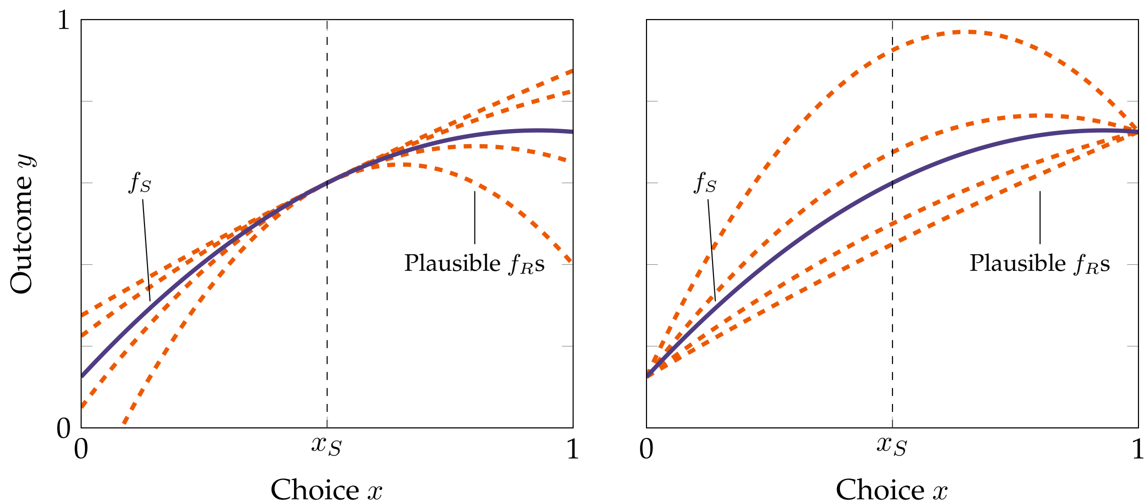
Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta(x - x_S)^2$ and $\mu_S = \mu_S(x_S, \sigma)$. Then, there exists a unique $\theta \ni \delta$ such that for all $f_R \in \phi(f_S; \theta, \mu_S)$, we have:*

$$f_R(x) = \beta_0 + \beta_1 x - \gamma(x - x_S)^2 + (\gamma - \delta)\sigma^2.$$

The set of models that R considers takes a simple form. There is a baseline "linear regression" component of each f_R that is the same as f_S: $\beta_0 + \beta_1 x$. There is then a quadratic component, centered around x_S (where γ is potentially unknown). Finally, there is a constant term that depends on γ, δ, and σ^2. Notably, if $\theta = [0, \overline{\gamma}]$, ϕ will be a signleton containing the true DGP. Otherwise, there may be multiple models that rationalize the data.

The intuition can be described as follows. First, we utilize a result known as "Stein's Lemma" that the derivative of f_R and f_S must be the same "on average" over μ_S. As $\sigma \to 0$, for each $f_R \in \phi(f_S; \theta, \mu_S)$, the derivative of f_R and f_S converges at the point x_S. Using (1), we also have that as $\sigma \to 0$, the value of each f_R and f_S must converge at f_S. Thus, all $f_R \in \phi(f_S; \theta, \mu_S)$ are (approximately) tangent to f_S at x_S.

This intuition is shown in the left panel of Figure 2 below. The solid line shows f_S, which has a curvature δ. However, given the sender's type θ may not account for all potential values that γ could take, there are other models f_R which could have generated the data, which are in turn reflected in the dashed blue lines. Crucially, all these lines are tangent to f_S at x_S.



(a) Plausible models when $\sigma = \underline{\sigma}$

(b) Plausible models when $\sigma > 0$

Figure 2: Plausible receiver models f_R as function of sender model f_S and sampling strategy $\mu_S(x_S, \sigma)$

The set of feasible models is different for $\sigma > 0$. In particular, if $\sigma' > \sigma$, then the distribution $\mu_S(x_S, \sigma')$ is a mean-preserving spread of $\mu_S(x_S, \sigma)$. Since f_R and f_S are both quadratic, their derivatives are linear, meaning (**??**) is invariant to mean preserving spreads (and thus does not depend on σ. Thus, each plausible f_R at σ is also plausible at σ' and vice-versa, up to a constant. The value of this constant depends on the curvature of f_R relative to f_S. Because each f_R is concave, its expectation over $\mu_S(x_S, \sigma)$ is decreasing in σ. If a given f_R is "more concave" than f_S — i.e. if $\gamma > \delta$, this f_R must shift up by a constant relative to f_S. If a given f_R is "less concave" than f_S — i.e. if $\gamma < \delta$, f_R must shift down by a constant relative to f_S. This is shown in the right panel of Figure 2. For higher σ, the plausible f_Rs that are "more linear" lie below f_S in the sampling range. The f_Rs that are "more concave" lie above f_S.

Receiver's Decisionmaking The receiver's learning takes a relatively simple form: regardless of the limitation on S's ability, R always learns what the data-generating-process looks like "up to the first derivative." Moreover, as σ changes — and thus S's sampling strategy changes — the set of plausible f_Rs remains identical, except that they shift up or down by a constant. This means that R's optimal *action* depends only her expectation of γ (given δ and θ). Given f_S, the expected value of the problem also has a simple expression. We denote For ease of notation, we will denote $\underline{\delta}^\theta = \min\{\theta\}$ and $\overline{\delta}^\theta = \max\{\theta\}$

Corollary 1. *R's optimal action is given by:*

$$x^* = x_S + \frac{\beta_1}{2\mathbb{E}_\gamma[\gamma|\theta, \delta]}.$$

R's expected value from this action is given by

$$y^* = \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\mathbb{E}_\gamma[\gamma|\theta,\delta]} + \big(\mathbb{E}_\gamma[\gamma|\theta,\delta] - \delta\big)\sigma^2.$$

The value of $\mathbb{E}_\gamma[\gamma|\theta,\delta]$ is also relatively simple.

- If (3) is equal to 0 at δ, then $\mathbb{E}_\gamma[\gamma|\theta,\delta] = \delta$.
- If (3) is ≤ 0, then it must be that $\delta = \overline{\delta}^\theta$, and $\mathbb{E}_\gamma[\gamma|\theta,\delta] = \mathbb{E}[\gamma|\gamma \geq \delta]$.
- Likewise, if (3) is ≥ 0, then it must be that $\delta = \underline{\delta}$, and $\mathbb{E}_\gamma[\gamma|\theta,\delta] = \mathbb{E}[\gamma|\gamma \leq \delta]$.

3.2 Sender's Data Sampling and Model Provision

We use these insights to derive the first result of the paper, which shows how S chooses a sampling strategy (i.e. σ) as a function of her type. We will use the following terminology to distinguish different types of sampling.

- **Local sampling**: $\sigma = \underline{\sigma}$. In this case, μ_S samples x values in a small neighborhood of x_S, i.e. is collecting information "locally relevant" to x_S.
- **Global sampling**: $\sigma = \overline{\sigma}$. In this case, μ_S samples x values that are farther and farther away from x_S, i.e. including information that potentially "irrelevant" (or at least spatially far) from x_S.

We also introduce the following notation

$$\overline{\gamma}^\theta = \mathbb{E}_\gamma[\gamma|\gamma \geq \overline{\delta}^\theta]$$
$$\underline{\gamma}^\theta = \mathbb{E}_\gamma[\gamma|\gamma \leq \underline{\delta}^\theta]$$

These represent R's expectation of γ, given noting that $\mathbb{E}_\gamma$ is taken with respect to R's beliefs.

We now state our result on sender sampling.

Theorem 1. *Supose $\theta = [\underline{\delta}^\theta, \overline{\delta}^\theta]$. Then, the optimal degree of sampling for a type θ sender σ_θ is equal to either $\overline{\sigma}$ or $\underline{\sigma}$. If $\overline{\delta}^\theta > \underline{\delta}^\theta$, S distantly samples if and only if*

$$p_\gamma^\theta(\overline{\delta}^\theta)(\overline{\gamma}^\theta - \overline{\delta}^\theta) \geq p_\gamma^\theta(\underline{\delta}^\theta)(\underline{\delta}^\theta - \underline{\gamma}^\theta),$$

in which case $\sigma_\theta = \overline{\sigma}$. Otherwise, $\sigma_\theta = \underline{\sigma}$, and S locally samples. Finally, if $\overline{\delta}^\theta = \underline{\delta}^\theta$, $\sigma_\theta = \overline{\sigma}$ when $\mathbb{E}_\gamma[\gamma] \geq \overline{\delta}^\theta$ and $\sigma_\theta = \overline{\sigma}$ when $\mathbb{E}_\gamma[\gamma] \leq \overline{\delta}^\theta$

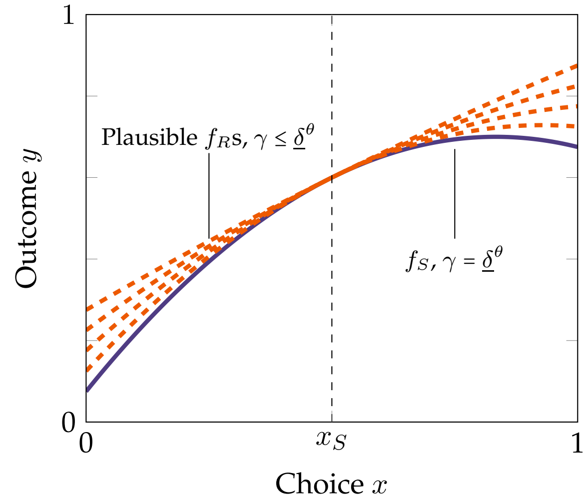
Theorem 1 can be interpreted as follows. If $\delta \in (\underline{\delta}^\theta, \overline{\delta}^\theta)$, then R believes S's models is correct. However, if $\delta = \overline{\delta}^\theta$, R will consider models with $\gamma \geq \overline{\delta}^\theta$. In this case, S's tends to estimate models that understate the magnitude of a potential tradeoff. This creates an incentive for him to estimate his model on information that is potentially irrelevant to learning about the DGP in a neighborhood of x_S.

However, when $\delta = \underline{\delta}^\theta$, the opposite happens: R considers models with $\gamma \leq \underline{\delta}^\theta$, i.e. "flatter" than those that S provided. This gives S an incentive to estimate the DGP "locally" in a neighborhood of x_S. If a type θ S tends to consider models with only high values of γ — i.e. to "overstate" the sharpness of a tradeoff — then he does have an incentive to locally sample choices in a neighborhood of x_S.

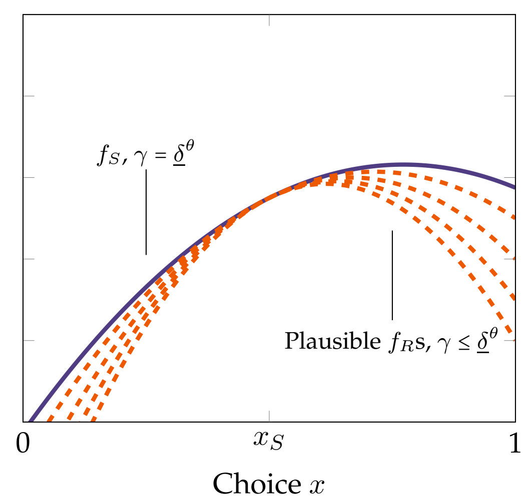
The intuition is most easily seen in Figure 3 below. The left panels (a) and (c) look at the case where f_S is such that $\delta = \underline{\delta}^\theta$, and thus where R entertains models that are "less steep" than f_S. The right panels (b) and (d) look at the case where f_S is such that $\delta = \overline{\delta}^\theta$, and R entertains steeper models (with higher γ).

In panels (a) and (b), we first look at the case where $\sigma = \underline{\sigma}$ and S locally samples around x_S. In both cases, the set of plausible models f_R is approximately tangent to f_S at x_S. However, in panel (a), because these f_Rs are 'shallower" (i.e. have lower γ values than $\underline{\delta}^\theta$), each function lies above f_S. By contrast, in panel (b), each f_R is "steeper" than f_S (i.e. has $\gamma \geq \overline{\delta}^\theta$), and thus lie *below* f_S.

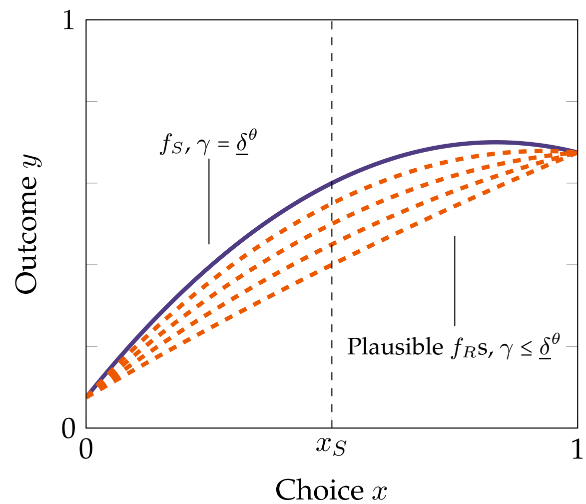
This logic flips when looking at panels (c) and (d). Recall that the distribution of x values when $\sigma = \overline{\sigma}$ is a mean-preserving-spread of that when $\sigma = \underline{\sigma}$. To rationalize the data, functions with higher γ values that are "more concave" must shift up relative to shallower ones. In panel (c), this means that the shallower f_R functions now lie *below* f_S. In panel (d), the steeper f_R functions now lie *above* f_S.



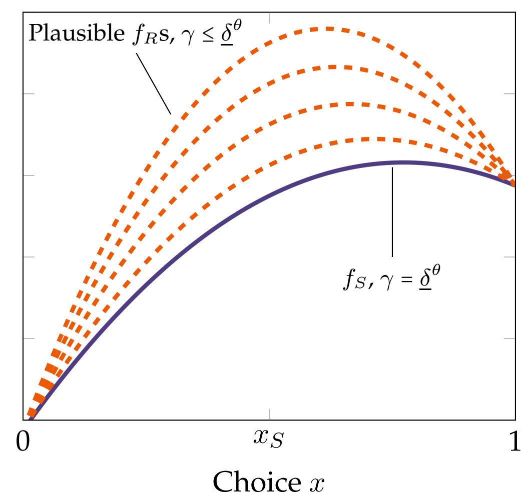
(a) Plausible f_Rs when $\sigma = \underline{\sigma}$, $\delta = \underline{\delta}^\theta$



(b) Plausible f_Rs when $\sigma = \underline{\sigma}$, $\delta = \overline{\delta}^\theta$



(c) Plausible f_Rs when $\sigma = \overline{\sigma}$, $\delta = \underline{\delta}^\theta$



(d) Plausible f_Rs when $\sigma = \overline{\sigma}$, $\delta = \overline{\delta}^\theta$

Figure 3: Models R entertains when $\delta = \underline{\delta}^\theta$ vs. $\delta = \overline{\delta}$, local sampling ($\sigma = \underline{\sigma}$) vs. distant sampling ($\sigma = \overline{\sigma}$)

Thus, whether $\sigma_\theta = \overline{\sigma}$ or $\underline{\sigma}$ depends on which scenario is more likely for S. If he is likely to end up in situations where the set of functions R will see is "shallower" than the "shallowest" function his type can model, he will prefer local sampling, as this effectively makes his model a lower bound for the set of models R entertains. If he is likely to end up in situations where the set of functions R will see is "steeper," he will prefer distant sampling.

In an extreme case, suppose $\theta = [\underline{\delta}^\theta, \overline{\gamma}]$. In this case, S can model the steepest tradeoffs R will entertain, but not shallower ones. Thus, he will always prefer local sampling. On the other hand, suppose $\theta = [0, \overline{\delta}^\theta]$. Then, any additional models that R entertains are "steeper" than f_S, so he has an incentive to distantly sample data and drive up R's expected value of these steeper problems.

Of particular interest is also the situation where $\theta = \{0\}$, which corresponds to a type of S who can only model *linear regressions*. Using Theorem 1, not only does such a sender always have an incentive to sample distantly, but as $\sigma \to \infty$, his perception of R's utility from his model also goes to ∞.

Corollary 2. *Suppose $\theta = \{0\}$, so that S only sends linear models. As $\sigma \to \infty$, $V(x_S) \to \infty$. S's perceptions about R's value also go to ∞.*

This logic is easily seen in the figure below: Fixing the sender's (linear regression) model, as σ increases, the set of models the receiver entertains shift higher and higher. While linear regression models are highly tractable to estimate, their simplicity is often justified as based on "local linear approximations" of nonlinear data-generating processes. However, a sender who could *provide* a linear model would instead want to include data from all across the policy space.

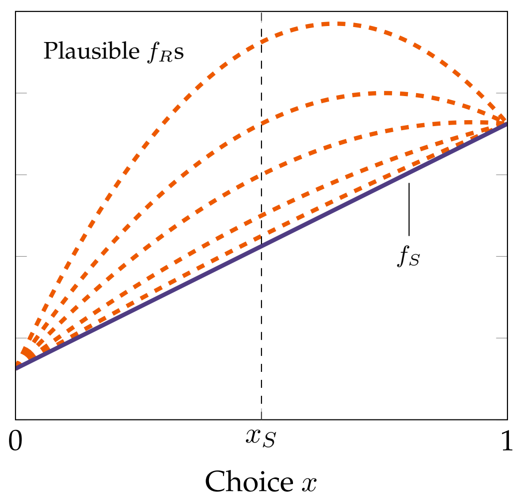


Figure 4: Plausible receiver models f_R when f_S is a linear regression

3.3 Receiver's Choice of x_S and Value of Information

In the absence of the sender, R would solely relyon her prior to choose x^*. This allows us to characterize the *value of information* for the sender.

Proposition 2. *The value of information for the receiver is given by*

$$V^I(x_S) = \frac{1}{4}\left(\int_\Theta \mathbb{E}_{\beta_1,\gamma}\left[\beta_1^2 q(\gamma,\theta)\right]dT(\theta) - \frac{\mathbb{E}_{\beta_1,\gamma}[\beta_1]^2}{\mathbb{E}_{\beta_1,\gamma}[\gamma]}\right)$$

where

$$q(\gamma,\theta) = \begin{cases} \frac{P_\gamma(\underline{\delta}^\theta)}{\underline{\gamma}^\theta} + \frac{P_\gamma(\overline{\delta}^\theta) - P_\gamma(\underline{\delta}^\theta)}{\gamma} + \frac{1-P_\gamma(\overline{\delta}^\theta)}{\overline{\gamma}^\theta} & \underline{\delta}^\theta < \overline{\delta}^\theta \\ \frac{1}{\mathbb{E}_\gamma[\gamma]} & \underline{\delta}^\theta = \overline{\delta}^\theta \end{cases}$$

Suppose a type θ sender entertains precisely the same models as the receiver — i.e. is such that $0 = \underline{\delta}^\theta < \overline{\delta}^\theta = \overline{\gamma}$. In this case, the value of information is proportional to $\mathbb{E}[\frac{\beta_1^2}{\gamma}] - \frac{\mathbb{E}_{\beta_1}[\beta_1]^2}{\mathbb{E}_\gamma[\gamma]}$. In this case, the receiver gains (relative to the prior) from both not only the local slope at x_S (β_1) but also the steepness of the curvature γ. As $\underline{\delta}^\theta$ increases or $\overline{\delta}^\theta$ decreases, the receiver learns less and less (in expectation) about γ. Finally, when $\underline{\delta}^\theta = \overline{\delta}^\theta$, the receiver effectively learns nothing about the curvature. However, she still retains value from hiring the sender, since she learns about the slope of f_R. In this final case, the value of information is proportional to $\frac{\mathrm{Var}_{\beta_1}(\beta_1)}{\mathbb{E}_\gamma[\gamma]}$.

Crucially, in the case of a receiver who maximizes $\mathbb{E}[y|x]$, the receiver's *expected utility* does not depend on the sender's choice of σ. The intuition is that different sampling strategies cause upwards and downwards shifts in the set of potential models the receiver entertains, which have no bearing on the *choice* the receiver eventually makes. Distant and local sampling will, however, affect the *variance* of the receiver's outcome, which we turn to next.

3.4 Forecast Accuracy

So far, we have assumed only that the receiver would like to maximize $\mathbb{E}[y|x]$. However, in many settings, the receiver may also directly (or indirectly) value *forecast accuracy*. A CEO, for instance, may not only value making the *right decision* about workforce restructuring, but may benefit from *accurately forecasting* the results of her restructuring. In fact, the less she is able to defend her choice with the "correct model," the more backlash she may face from the board of directors or employees at her firm. A receiver who values *forecast accuracy* intuitively prefers *model certainty*, which we have abstracted from so far.

To illustrate the impact of model simplicity and sampling on forecast-minded receivers, we consider a receiver whose expected utility is given by

$$(1-\lambda)\mathbb{E}[y|x] - \lambda \mathrm{Var}(y|x)$$

for $\lambda \in [0,1]$. Intuitively, these preferences can also be thought of as follows. In a first period, the receiver forms an expectation of y. In a second period, given this expectation, the utility she reaps is based on her ability to forecast. λ measures her patience; if $\lambda = 0$, she does not value the future or forecasting, and only cares about her expectation of y. As $\lambda \to 1$, she places a greater and greater emphasis on forecasting. We write a forecast-minded receiver's value as

$$V^F(f_S;\theta,\sigma,\lambda) = \max_x (1-\lambda)\mathbb{E}_{f_S;\theta,\sigma}[y|x] - \lambda \mathrm{Var}_{f_S;\theta,\sigma}[y|x].$$

Proposition 3. *Fix* $f_S = \beta_0 + \beta_1 x - \delta(x-x_S)^2$. *If* $x^* = \arg\max \mathbb{E}_{f_S;\theta,\sigma}[y|x]$, *then* $\mathrm{Var}_{f_S;\theta,\sigma}[y|x^*]$ *is decreasing in* σ^2. *Moreover, there exists* $\overline{\lambda} > 0$ *such that*

- *if* $\delta = \overline{\delta}^\theta < \overline{\gamma}$, *or*
- *if* $\delta = \underline{\delta}^\theta > 0$ *and* $\lambda \geq \overline{\lambda}$,

there exists $\sigma_V^* > 0$ such that $V^F(f_S;\theta,\sigma,\lambda)$ is maximized at $\sigma = \sigma_V^*$.

The intuition for the result is as follows. As σ increases, the set of models the receiver entertains shift upward or donwards. While the receiver's actual choice of x does not change, the expected value of y given x spreads out. Thus, $\text{Var}_{f_S;\theta,\sigma}[y|x]$ is increasing in σ^2.

This means that, if the receiver were simply to choose x^* — which would maximize $\mathbb{E}[y|x]$, distant sampling would be *worse* for the receiver's utility. Combined with Theorem 1, this suggests that a receiver is better off when facing a sender whose models tend to *overstate* tradeoffs — who have an incentive to distantly sample — than those whose models tend to *understate* tradeoffs — who *locally* sample.

However, what is less intuitive is that, even in the presence of forecast error, a sender may still have an incentive to distantly sample. The logic is shown in the figure below. Using (1), we show that at the points $x_S \pm \sigma$, each of the models the receiver entertains are equal to the sender's model. At this point the value of y given x is the same regardless of the true model, and $\text{Var}(y) = 0$. This value of y is precisely maximized when the receiver's models cross the sender's at the point $x_S + \sigma$.

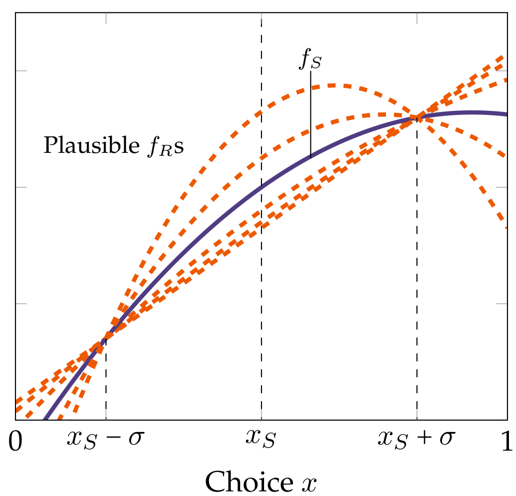


Figure 5: Intersection of receiver's models with sender's models

Thus, even if the receiver places vanishingly small weight on $\mathbb{E}[y|x]$ and values forecasting error, the sender still has an incentive to depart from local sampling. Moreover, unlike in the baseline case, there may be an incentive for the sender to sample more distantly if the receiver values forecast error.

3.5 Generalized Model

This section is under construction! Please click the link to see if this is available in a future draft. Argument outline:

- Consider convex loss function $\gamma \cdot c(\cdot)$

- Fixing γ, establish equivalency class: $\beta_0 + \beta_1 x - \gamma \cdot c(\cdot) \cong \alpha_0 + \alpha_1 x - \gamma \cdot c(\cdot)$. These same weight prior.

- Under mean preserving spreads: if "load" change in moment condition (2) onto β_1 or α_1, show that value of all functions increases at x_S. So things "shift up," although slope at x_S might go up or down.

- Theorem: if $\overline{\sigma}$ is sufficiently large, sampling with $\overline{\sigma}$ dominates $\underline{\sigma}$.

4 Gains from Simplification

So far, we have taken model oversimplification as an exogenous feature of our informational environment. This section studies when a sender benefits from simplification, thus providing an *endogenous* channel for the sender to provide simplified models. First, we characterize *when* different types of senders benefit from providing a simplified model. Then, we consider the problem of a "type space designer," who can endogenously construct a type space Θ. Senders are randomly drawn from this type space to maximize the receiver's expectations of y.

We show that if this designer considers exactly the same models the receiver, the optimal design of the type space is composed of *singletons*. In particular, for every $\gamma \in [0, \overline{\gamma}]$, there is a type $\theta = \{\gamma\}$. Thus, sender types each have a fixed (but arbitrary) of modeling the receiver's tradeoff, meaning the receiver *never* learns the true value of γ.

4.1 Characterizing Benefits from Oversimplification

Previously, we assumed that senders differed in their ability to model the quadratic component of the receiver's models, that different receivers could model this component in distinct ways, and thus that the receiver could fully back out the sender's type from the model she received. However, we now relax Assumption 3 and allow the sender's types to have overlapping supports. Let $\theta_i, \theta_j \in \Theta$. With some abuse of notation, we write

$$\theta_i = [\underline{\delta}^i, \overline{\delta}^i]$$
$$\theta_j = [\underline{\delta}^j, \overline{\delta}^j]$$
$$\theta_i \cap \theta_j \neq \varnothing.$$

Let $\delta \in \theta_i \cap \theta_j$. Conditional on seeing a model of the form $y = \beta_0 + \beta_1 x - \delta(x - x_S)^2$, the receiver does not know whether this model came from a type θ_i or type θ_j sender. In particular, we can characterize when uncertainty about the source of the signal can actually work in a sender's benefit, and when it hurts the sender.

Proposition 4. *Suppose $\theta_i \cap \theta_j \neq \varnothing$ and $\theta_i^\circ, \theta_j^\circ \neq \varnothing$. Let $\sigma_i, \sigma_j \in [\underline{\sigma}, \overline{\sigma}]$ of the type i, j senders. Then, the type i sender* strictly *gains (loses) from type j's existence when*

- $\underline{\delta}_i \in \theta_j^{\circ}$ *and* $\sigma_i = \overline{\sigma}$ $(\sigma_i = \underline{\sigma})$;
- $\overline{\delta}_i \in \theta_j^{\circ}$ *and* $\sigma_i = \underline{\sigma}$ $(\sigma_i = \overline{\sigma})$;
- $\delta \in \theta_i^{\circ}$, $\delta = \overline{\delta}_j$, *and* $\sigma_j = \overline{\sigma}$ $(\sigma_j = \underline{\sigma})$;
- $\delta \in \theta_i^{\circ}$, $\delta = \underline{\delta}_j$, *and* $\sigma_j = \underline{\sigma}$ $(\sigma_j = \overline{\sigma})$;
- $\overline{\delta}_i = \underline{\delta_j}$, $\sigma_i = \underline{\sigma}$, *and* $\sigma_j = \overline{\sigma}$ $(\sigma_i = \overline{\sigma}$ *and* $\sigma_j = \underline{\sigma})$;
- *and* $\underline{\delta}_i = \overline{\delta_j}$, $\sigma_i = \overline{\sigma}$, *and* $\sigma_j = \underline{\sigma}$ $(\sigma_i = \underline{\sigma}$ *and* $\sigma_j = \overline{\sigma})$.

The intuition

Corollary 3. *If one guy is better off the other is worse off.*

4.2 Optimal Type Space Design

This section is under construction! Please click the link to see if this is available in a future draft.

5 Competition

This section is under construction! Please click the link to see if this is available in a future draft.

6 Conclusion

This section is under construction! Please click the link to see if this is available in a future draft.

A Proofs of Results

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta(x - x_S)^2$ and $\mu_S = \mu_S(x_S, \sigma)$. Then, there exists a unique $\theta \ni \delta$ such that for all $f_R \in \phi(f_S; \theta, \mu_S)$, we have:*

$$f_R(x) = \beta_0 + \beta_1 x - \gamma(x - x_S)^2 + (\gamma - \delta)\sigma^2.$$

Proof. Because sender type intervals θ are non-overlapping, the value of δ in $f_S(x) = \beta_0 + \beta_1 - \delta(x - x_S)^2$ uniquely pins down the sender's type θ.

Suppose $f_R \in \phi(f_S; \theta, \mu_S)$. Rewriting f_R by expanding the Taylor Series around x_S and adding x_S, we write f_R as

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma(x - x_S)^2 + k,$$

where k is a constant. Because μ_S is symmetric, it is elliptical (in one dimension). By Stein's Lemma (Landsman and Nešlehová, 2008), equation (2) can be written as

$$\begin{aligned}
\mathbb{E}_{\mu_S}\big[x\big(f_R(x) - f_S(x)\big)\big] &= \mathrm{Cov}_{\mu_S}(x, f_R(x) - f_S(x)) \\
&= \sigma^2 \mathbb{E}_{\mu_S}[f_R'(x) - f_S'(x)] \\
&= \sigma^2 \mathbb{E}_{\mu_S}[\alpha_1 - 2\delta(x - x_S) - \beta_1 + 2\gamma(x - x_S)] = 0
\end{aligned}$$

Because the mean of μ_S is x_S, this yields the condition $\alpha_1 = \beta_1$ which is *independent* of σ^2. Hence, we have

$$f_R(x) = \beta_0 + \beta_1 x - \gamma(x - x_S)^2 + k$$

with the value of γ and k potentially indeterminate. However, fixing a plausible γ value, from (1) we have

$$\begin{aligned}
\mathbb{E}_{\mu_S}[f_R(x)] = \beta_0 + \beta_1 x_S - \gamma \mathbb{E}_{\mu_S}[(x - x_S)^2] + k = \beta_0 + \beta_1 x_S - \delta \mathbb{E}_{\mu_S}[(x - x_S)^2] &= \mathbb{E}_{\mu_S}[f_S(x)] \\
\implies -\gamma\sigma^2 + k &= -\delta\sigma^2 \\
\implies k &= (\gamma - \delta)\sigma^2.
\end{aligned}$$

completing the proof. □

Theorem 1. *Supose $\theta = [\underline{\delta}^\theta, \overline{\delta}^\theta]$. Then, the optimal degree of sampling for a type θ sender σ_θ is equal to either $\overline{\sigma}$ or $\underline{\sigma}$. If $\overline{\delta}^\theta > \underline{\delta}^\theta$, S distantly samples if and only if*

$$p_\gamma^\theta(\overline{\delta}^\theta)(\overline{\gamma}^\theta - \overline{\delta}^\theta) \geq p_\gamma^\theta(\underline{\delta}^\theta)(\underline{\delta}^\theta - \underline{\gamma}^\theta),$$

in which case $\sigma_\theta = \overline{\sigma}$. Otherwise, $\sigma_\theta = \underline{\sigma}$, and S locally samples. Finally, if $\overline{\delta}^\theta = \underline{\delta}^\theta$, $\sigma_\theta = \overline{\sigma}$ when $\mathbb{E}_\gamma[\gamma] \geq \overline{\delta}^\theta$ and $\sigma_\theta = \overline{\sigma}$ when $\mathbb{E}_\gamma[\gamma] \leq \overline{\delta}^\theta$

Proof. Following from the Proposition 1, let $\theta = [\underline{\delta}^\theta, \overline{\delta}^\theta]$. Suppose $\theta^\circ \neq \varnothing$ and $\delta \in \theta^\circ$. Then (3) is necessarily satisfied, and $\phi(f_S; \theta, \mu_S) = \{f_S\}$. However, if $\theta = \underline{\theta}$ or $\overline{\theta}$, then (3) is a corner solution from the receiver's perspective. If $\delta = \overline{\delta}^\theta$, then any $\gamma \geq \overline{\delta}^\theta$ is plausible from the receiver's perspective. If $\delta = \underline{\delta}^\theta$, any $\gamma \leq \underline{\delta}^\theta$ is plausible. Finally, if $\underline{\delta}^\theta = \overline{\delta}^\theta$, any $\gamma \in [0, \overline{\gamma}]$ value is possible. This pins down the plausible set of γ values from the receiver's perspective.

Next, fix σ. The receiver chooses x to maximize:

$$\beta_0 + \beta_1 x - \mathbb{E}[\gamma|\delta](x - x_S)^2 + \big(\mathbb{E}[\gamma|\delta] - \delta\big)\sigma^2.$$

The receiver's choice is given by $x^* = x_S + \frac{\beta_1}{2\mathbb{E}[\gamma|\delta]}$. In this case, her expected value is:

$$\beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\mathbb{E}[\gamma|\delta]} + (\mathbb{E}[\gamma|\delta] - \delta)\sigma^2.$$

Suppose $\delta \in \theta^\circ$. Then, $\mathbb{E}[\gamma|\delta] = \delta$, and the receiver's expected value does not depend on σ^2. If $\delta = \overline{\delta}^\theta$, her expected utility is

$$\beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\overline{\gamma}^\theta} + (\overline{\gamma}^\theta - \delta)\sigma^2,$$

recalling that $\overline{\gamma}^\theta = \mathbb{E}[\gamma|\gamma \geq \overline{\delta}^\theta]$. Similarly, we have that if $\delta = \underline{\delta}^\theta$, her expected utility is

$$\beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\underline{\gamma}^\theta} + (\underline{\gamma}^\theta - \delta)\sigma^2.$$

Working backwards, S then chooses σ^2 to maximize

$$p_\gamma^\theta(\overline{\delta}^\theta)\left(\frac{\mathbb{E}[\beta_1^2]}{4\overline{\gamma}^\theta} + (\overline{\gamma}^\theta - \overline{\delta}^\theta)\sigma^2\right) + p_\gamma^\theta(\underline{\delta}^\theta)\left(\frac{\mathbb{E}[\beta_1^2]}{4\underline{\gamma}^\theta} + (\underline{\gamma}^\theta - \underline{\delta}^\theta)\sigma^2\right).$$

The derivative with respect to σ^2 is simply

$$p_\gamma^\theta(\overline{\delta}^\theta)(\overline{\gamma}^\theta - \overline{\delta}^\theta) - p_\gamma^\theta(\underline{\delta}^\theta)(\underline{\delta}^\theta - \underline{\gamma}^\theta).$$

If the former is greater than the latter, the optimal choice is $\sigma = \overline{\sigma}$. Otherwise, $\sigma = \underline{\sigma}$. This gives the result. Finally, in the case where θ is degenerate and equal to $\{\delta\}$, the preference for distant sampling reduces to:

$$\mathbb{E}[\gamma] - \delta \geq 0.$$

□

Proposition 2. *The value of information for the receiver is given by*

$$V^I(x_S) = \frac{1}{4}\bigg(\int_\Theta \mathbb{E}_{\beta_1,\gamma}\big[\beta_1^2 q(\gamma,\theta)\big] dT(\theta) - \frac{\mathbb{E}_{\beta_1,\gamma}[\beta_1]^2}{\mathbb{E}_{\beta_1,\gamma}[\gamma]}\bigg)$$

where

$$q(\gamma,\theta) = \begin{cases} \frac{P_\gamma(\underline{\delta}^\theta)}{\underline{\gamma}^\theta} + \frac{P_\gamma(\overline{\delta}^\theta) - P_\gamma(\underline{\delta}^\theta)}{\gamma} + \frac{1-P_\gamma(\overline{\delta}^\theta)}{\overline{\gamma}^\theta} & \underline{\delta}^\theta < \overline{\delta}^\theta \\ \frac{1}{\mathbb{E}_\gamma[\gamma]} & \underline{\delta}^\theta = \overline{\delta}^\theta \end{cases}$$

Proof. Given a type $\theta = [\underline{\delta}^\theta, \overline{\delta}^\theta]$ sender, a sender model $\beta_0 + \beta_1 x - \delta(x - x_S)^2$, and a sampling strategy σ_θ, R's value is given by

$$V(x_S|\theta,\delta,\sigma_\theta) = \begin{cases} \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\delta} & \delta \in (\underline{\delta}^\theta, \overline{\delta}^\theta) \\ \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\overline{\gamma}^\theta} + (\overline{\gamma}^\theta - \overline{\delta}^\theta)\sigma_\theta^2 & \delta \geq \overline{\delta}^\theta \\ \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\underline{\gamma}^\theta} + (\underline{\gamma}^\theta - \underline{\delta}^\theta)\sigma_\theta^2 & \delta \leq \underline{\delta}^\theta \end{cases}$$

The receiver then integrates $V(x_S|\theta,\delta,\sigma_\theta)$ over the potential δ values, yielding

$$\begin{aligned} V(x_S|\theta,\sigma_\theta) = {} & \overbrace{\beta_0 + \sigma_\theta^2\big[P_\gamma(\underline{\delta}^\theta)(\underline{\gamma}^\theta - \underline{\delta}^\theta) + (1 - P_\gamma(\overline{\delta}^\theta))(\overline{\gamma}^\theta - \overline{\delta}^\theta)\big]}^{\text{Constant term}} + \beta_1 x_S \\ & + \frac{\beta_1^2}{4}\Big[\frac{P_\gamma(\underline{\delta}^\theta)}{\underline{\gamma}^\theta} + \frac{P_\gamma(\overline{\delta}^\theta) - P_\gamma(\underline{\delta}^\theta)}{\delta} + \frac{1 - P_\gamma(\overline{\delta}^\theta)}{\overline{\gamma}^\theta}\Big] \end{aligned}$$

Note that, given the receiver's models take the form $y = \alpha_0 + \alpha_1 x - \gamma(x - x_S)^2$, note that the "constant term" in the expression above is simply a noisy signal about α_0. Because $\mathbb{E}[y]$ is linear in α_0 and the receiver is risk-neutral in y, ex-ante, we have that

$$\mathbb{E}[\alpha_0] = \mathbb{E}\big[\beta_0 + \sigma_\theta^2\big[P_\gamma(\underline{\delta}^\theta)(\underline{\gamma}^\theta - \underline{\delta}^\theta) + (1 - P_\gamma(\overline{\delta}^\theta))(\overline{\gamma}^\theta - \overline{\delta}^\theta)\big]\big].$$

Since α_0 Similarly, $\mathbb{E}[\beta_1]x_S$ is linear in beliefs about β_1. This means we can calculate the value of information $\mathbb{E}[V_S(x_S|\theta,\sigma_\theta)] - \mathbb{E}[V(x_S)]$ as:

$$\mathbb{E}_{\beta_1,\gamma}\bigg[\frac{\beta_1^2}{4}\Big[\frac{P_\gamma(\underline{\delta}^\theta)}{\underline{\gamma}^\theta} + \frac{P_\gamma(\overline{\delta}^\theta) - P_\gamma(\underline{\delta}^\theta)}{\gamma} + \frac{1 - P_\gamma(\overline{\delta}^\theta)}{\overline{\gamma}^\theta}\bigg] - \frac{\mathbb{E}_{\beta_1,\gamma}[\beta_1]^2}{4\mathbb{E}_{\beta_1,\gamma}[\gamma]}$$

for $\underline{\delta}^\theta < \overline{\delta}^\theta$. In the case where $\underline{\delta}^\theta = \overline{\delta}^\theta$, this reduces to

$$\frac{\mathbb{E}_{\beta_1,\gamma}[\beta_1^2] - \mathbb{E}_{\beta_1,\gamma}[\beta_1]^2}{4\mathbb{E}_{\beta_1,\gamma}[\gamma]}.$$

Thus, define the function

$$q(\gamma,\theta) = \begin{cases} \frac{P_\gamma(\underline{\delta}^\theta)}{\underline{\gamma}^\theta} + \frac{P_\gamma(\overline{\delta}^\theta) - P_\gamma(\underline{\delta}^\theta)}{\gamma} + \frac{1-P_\gamma(\overline{\delta}^\theta)}{\overline{\gamma}^\theta} & \underline{\delta}^\theta < \overline{\delta}^\theta \\ \frac{1}{\mathbb{E}_\gamma[\gamma]} & \underline{\delta}^\theta = \overline{\delta}^\theta \end{cases}$$

Then, the ex-ante value of information, conditional on not knowing the type of the sender, is proportional to

$$\int_\Theta \mathbb{E}_{\beta_1,\gamma}\big[\beta_1^2 q(\gamma,\theta)\big] dT(\theta) - \frac{\mathbb{E}_{\beta_1,\gamma}[\beta_1]^2}{\mathbb{E}_{\beta_1,\gamma}[\gamma]}.$$

□

Proposition 3. *Fix* $f_S = \beta_0 + \beta_1 x - \delta(x - x_S)^2$. *If* $x^* = \arg\max \mathbb{E}_{f_S;\theta,\sigma}[y|x]$, *then* $\mathrm{Var}_{f_S;\theta,\sigma}[y|x^*]$ *is decreasing in* σ^2. *Moreover, there exists* $\overline{\lambda} > 0$ *such that*

- *if* $\delta = \overline{\delta}^\theta < \overline{\gamma}$, *or*
- *if* $\delta = \underline{\delta}^\theta > 0$ *and* $\lambda \geq \overline{\lambda}$,

there exists $\sigma_V^* > 0$ *such that* $V^F(f_S;\theta,\sigma,\lambda)$ *is maximized at* $\sigma = \sigma_V^*$.

Proof. Let x^* maximize $\mathbb{E}[y|x]$. Given the sender's model, the receiver's only uncertainty is over γ. This means that

$$\mathrm{Var}_{f_S;\theta,\sigma}(y|x^*) = \mathrm{Var}_{\gamma;\sigma,\theta} = \begin{cases} 0 & \gamma = \delta \in (\underline{\delta}^\theta, \overline{\delta}^\theta) \\ \mathrm{Var}_{\gamma \geq \overline{\delta}^\theta}\big((\frac{\beta_1^2}{4\gamma} + (\gamma - \overline{\delta}^\theta)\sigma_\theta^2)\big) & \gamma \geq \overline{\delta}^\theta \\ \mathrm{Var}_{\gamma \leq \underline{\delta}^\theta}\big((\frac{\beta_1^2}{4\gamma} + (\gamma - \overline{\delta}^\theta)\sigma_\theta^2)\big) & \gamma \geq \underline{\delta}^\theta \end{cases}$$

In the case where $\gamma \geq \overline{\delta}^\theta$, this can be simplified as

$$\frac{\beta_1^4}{16}\mathrm{Var}_{\gamma \geq \overline{\delta}^\theta}\Big(\frac{1}{\gamma}\Big) + \sigma_\theta^4 \mathrm{Var}_{\gamma \leq \overline{\delta}^\theta}(\gamma),$$

which is increasing in σ. A similar condition holds for the case where $\gamma \leq \underline{\delta}^\theta$. This gives the first part of the result.

For the second part of the result, note that

$$\mathrm{Var}_{f_S;\theta,\sigma}[y|x] = [\sigma^2 - (x - x_S)^2]^2 \mathrm{Var}_\gamma(\gamma).$$

Thus, when $x = x_S \pm \sigma$, the variance of outcomes equals zero. Now, letting x^* denote the maximizer associated with $V^F(f_S;\theta,\sigma,\lambda)$, we have:

$$\frac{\partial}{\partial\sigma^2}\max_x(1-\lambda)\mathbb{E}_{f_S;\theta,\sigma}[y|x]-\lambda\mathrm{Var}_{f_S;\theta,\sigma}[y|x]=(1-\lambda)(\mathbb{E}_\gamma[\gamma]-\delta)-2\lambda[\sigma^2-(x^*-x_S)^2]\mathrm{Var}_\gamma(\gamma) \tag{4}$$

If $\delta\in\theta^\circ$, this expression is 0. So, first suppose $\delta=\overline{\delta}^\theta<\overline{\gamma}$. In this case, whenever $\sigma^2\leq(x^*-x_S)^2$, both sets of terms are positive. Thus, there exists $\sigma_V^*>0$ that maximizes the expression.

If $\delta=\underline{\delta}^\theta>0$, the first term is always negative. The second term is positive at $\sigma^2=0$. If $\lambda\to1$, the second term outweighs the first, and the derivative is positive at $\sigma^2=0$. More precisely, x^* solves

$$(1-\lambda)\big(\beta_1-2\mathbb{E}_\gamma[\gamma](x^*-x_S)\big)+4\lambda(\sigma^2-(x^*-x_S)^2)(x^*-x_S)\mathrm{Var}_\gamma(\gamma)=0.$$

Let $z=x^*-x_S$. We have

$$(1-\lambda)\big(\beta_1-2\mathbb{E}_\gamma[\gamma]z\big)+4\lambda(\sigma^2-z^2)z\mathrm{Var}_\gamma(\gamma)=0.$$

Letting z' denote the derivative of z with respect to σ^2, we then have

$$\begin{aligned}-2(1-\lambda)\mathbb{E}_\gamma[\gamma]z'+4\lambda\sigma^2\mathrm{Var}_\gamma(\gamma)z'+4\lambda z\mathrm{Var}_\gamma(\gamma)-12\lambda z^2\mathrm{Var}_\gamma(\gamma)z'=0\\ \Longrightarrow z'=\frac{4\lambda z\mathrm{Var}_\gamma(\gamma)}{2(1-\lambda)\mathbb{E}_\gamma[\gamma]+12\lambda z^2\mathrm{Var}_\gamma(\gamma)-4\lambda\sigma^2\mathrm{Var}_\gamma(\gamma)}.\end{aligned}$$

Differentiating (4) with respect to σ^2 gives

$$-2\lambda\mathrm{Var}_\gamma(\gamma)+4\lambda zz'\mathrm{Var}_\gamma(\gamma)\geq0\iff zz'\geq\frac{1}{2}\iff\frac{4\lambda z^2\mathrm{Var}_\gamma(\gamma)}{(1-\lambda)\mathbb{E}_\gamma[\gamma]+2\lambda(3z^2-\sigma^2)\mathrm{Var}_\gamma(\gamma)}\geq1.$$

□

Proposition 4. *Suppose $\theta_i\cap\theta_j\neq\varnothing$ and $\theta_i^\circ,\theta_j^\circ\neq\varnothing$. Let $\sigma_i,\sigma_j\in[\underline{\sigma},\overline{\sigma}]$ of the type i,j senders. Then, the type i sender* strictly *gains (loses) from type j's existence when*

- *$\underline{\delta}_i\in\theta_j^\circ$ and $\sigma_i=\overline{\sigma}$ ($\sigma_i=\underline{\sigma}$);*
- *$\overline{\delta}_i\in\theta_j^\circ$ and $\sigma_i=\underline{\sigma}$ ($\sigma_i=\overline{\sigma}$);*
- *$\delta\in\theta_i^\circ$, $\delta=\overline{\delta}_j$, and $\sigma_j=\overline{\sigma}$ ($\sigma_j=\underline{\sigma}$);*
- *$\delta\in\theta_i^\circ$, $\delta=\underline{\delta}_j$, and $\sigma_j=\underline{\sigma}$ ($\sigma_j=\overline{\sigma}$);*
- *$\overline{\delta}_i=\underline{\delta}_j$, $\sigma_i=\underline{\sigma}$, and $\sigma_j=\overline{\sigma}$ ($\sigma_i=\overline{\sigma}$ and $\sigma_j=\underline{\sigma}$);*
- *and $\underline{\delta}_i=\overline{\delta}_j$, $\sigma_i=\overline{\sigma}$, and $\sigma_j=\underline{\sigma}$ ($\sigma_i=\underline{\sigma}$ and $\sigma_j=\overline{\sigma}$).*

Proof. First, we show how the receiver's expected value from entertaining models that have steeper or shallower tradeoffs than the sender's affect her maximal expected utility. Suppose $\gamma>\delta$. Given a sampling strategy σ, we have:

$$\max \beta_0 + \beta_1 x - \delta(x - x_S)^2 = \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\delta}$$
$$\max \beta_0 + \beta_1 x - \gamma(x - x_S)^2 + (\gamma - \delta)\sigma^2 = \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2.$$

The latter is greater than the former when

$$\frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4\delta} \iff (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4}\left(\frac{1}{\delta} - \frac{1}{\gamma}\right) \iff \sigma^2 \geq \frac{\beta_1^2}{\gamma\delta}.$$

(Make sure to make policy space bounded in right way).

Now, suppose $\delta \in \theta_i^\circ, \theta_j^\circ$. Then, the receiver entertains just the sender's model, which she believes is the sender's (only) model. Thus, we consider situations where the boundaries of the sets intersect.

Next, suppose $\underline{\delta}_i \in \theta_j^\circ$. Conditional on seeing $\underline{\delta}_i$, the receiver entertains models with $\gamma \leq \underline{\delta}_i$. By Theorem 1, the sender would receive to locally sample in these cases. However, in her initial choice of σ, the sender is forced to consider both the probability of $\overline{\delta}_i$ and $\underline{\delta}_i$. If $\sigma_i = \overline{\sigma}$, the sender is being forced to distantly sample, which delivers a gain most of the time but a loss in the case where $\underline{\delta}_i$ materializes. However, if $\underline{\delta}_i \in \theta_j^\circ$, then with some probability, the receiver believes the $\underline{\delta}_i$ model was the true model and came from the type j sender. This means the receiver places a higher weight on the sender's model itself, and less on models with lower γ values (which, under $\sigma_i = \overline{\sigma}$, deliver lower expected utility).

A similar argument shows that the sender *loses out* when $\delta = \underline{\delta} - i$ and $\sigma_i = \underline{\sigma}$. In this case, the sender is (optimally) gaining by herself locally sampling, meaning the remaining models the receiver entertains deliver higher expected utility. However, the receiver thinks with positive probability that the sender's model is in fact the *true* model, causing her to relatively downweight these alternative models and focus more weight on the sender's model, which delivers lower expected utility. Yet another similar arugment shows that the sender gains (loses) when $\overline{\delta}_i \in \theta_j^\circ$ and $\sigma_i = \underline{\sigma}$ ($\sigma_i = \overline{\sigma}$).

Next, suppose $\delta \in \theta_i^\circ$ and that $\delta = \overline{\delta}_j$. If the type j sender is distantly sampling ($\sigma_j = \overline{\sigma}$), the receiver entertains not only type i's model but also models with higher γ (since she believes the model could have also come from θ_j). The set of models R entertains is thus rosier than if she were to just entertain the type i sender's model, meaning the type i sender benefits from the type j's distant sampling Local sampling, on the other hand, would hurt her. A similar argument takes cares of the case where $\delta = \underline{\delta}_j$, where the type θ_i sender is hurt by distant sampling and benefits from local sampling.

$\square$

References

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